Bio-Unix & Shell Scripting

Introduction

prepared by

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UNIX

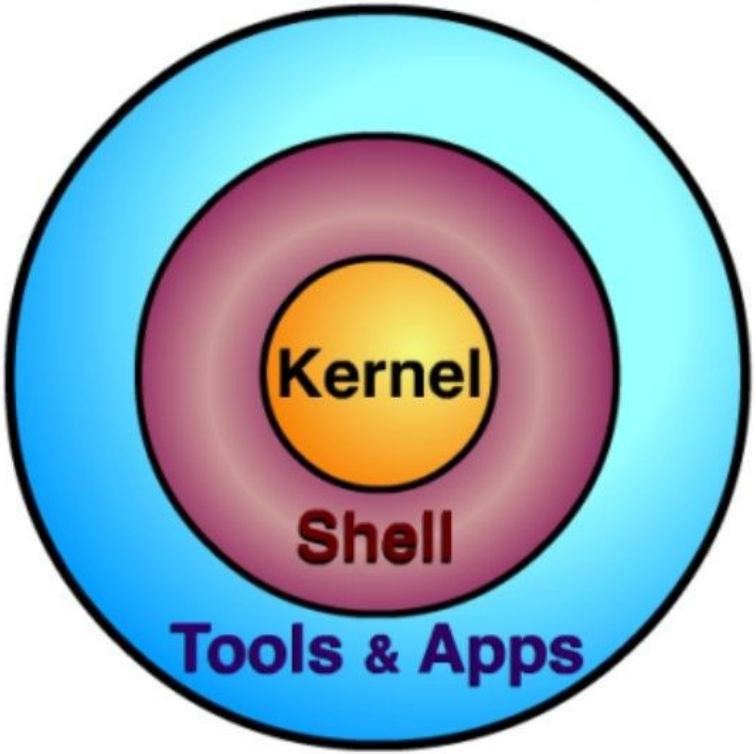
- Unix is an operating system (OS) developed in the late 1960s at AT&T's Bell Labs
- Used in various applications, including workstations, servers, and embedded systems
- Unix systems have both the Command Line Interface (CLI) and Graphical User Interface (GUI)
- There are many different varieties of UNIX and the most popular ones are Linux, Solaris, and MacOS



American Telephone and Telegraph Company

UNIX

Parts of a Unix OS



UNIX

- Kernel - The core/heart of the os. Manages processes, memory, software and hardware
- Shell - The shell serves as the interface between the user and kernel. Enables a user to instruct the system to perform tasks
- There are different types of shells like the bourne shell (sh), Bash (Bourne Again SHell), korn shell (ksh), C shell (csh), and the z shell (zsh)
- Tools and apps - Performs specific tasks for the user

LINUX 

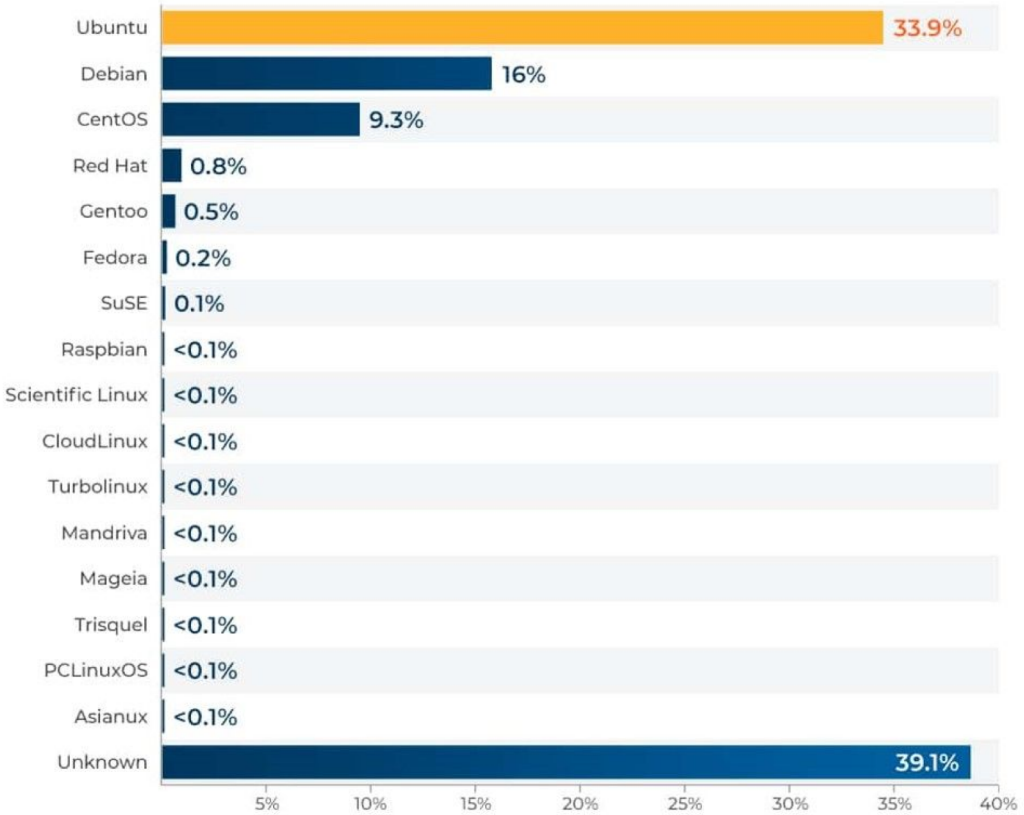
- A unix-like operating system developed by by Linus Torvalds in 1991
- Linux is free and open source whereas some of the original derivatives of Unix are not
- It is fast, secure, compatible with different platforms, multi-user and multi-tasking OS
- Linux has multiple distributions (distros) like Ubuntu, Debian, CentOS, Fedora, and others (https://distrowatch.com/)

Ubuntu 

- One of the most common and popular linux distributions
- Easy and convenient to use for most beginners
- Download and Installation instructions

  (https://ubuntu.com/tutorials/install-ubuntu-desktop#1-overview)

Ubuntu



Source: https://truelist.co/blog/linux-statistics/

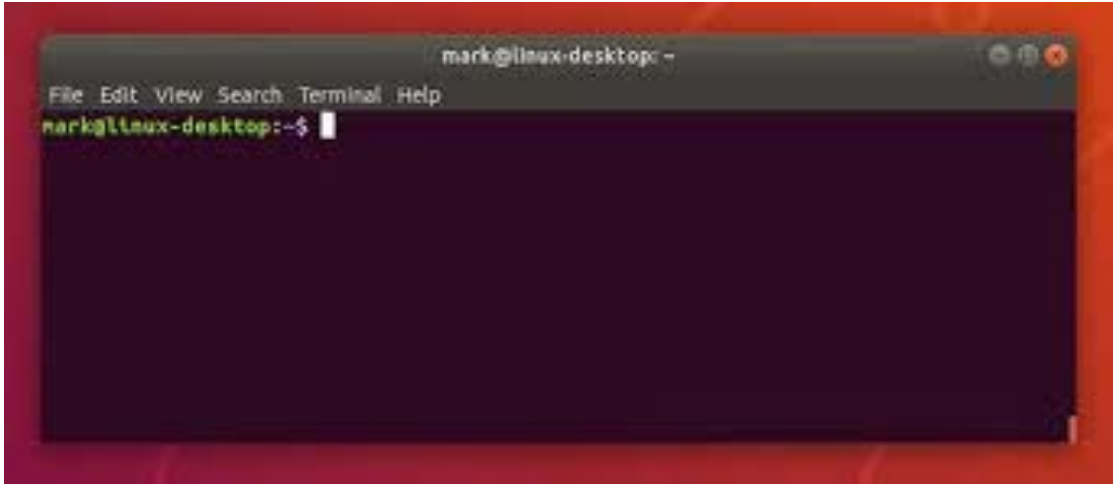
Windows Subsystem for Linux (WSL)

- WSL enables you to install a Linux distribution (Ubuntu, Kali, Debian, etc) and use Linux applications directly on windows
- This eliminates the traditional methods like using virtual machines and dual-booting
- Installation instructions

  (https://learn.microsoft.com/en-us/windows/wsl/install)

The terminal

- The terminal is a wrapper program that runs the shell
- The most common shell is bash and it comes as the default shell in most linux distributions



Applications of Unix Derivates and Scripting

- File management and manipulation
- Data filtering
- Performance boosting
- User and group management
- Automation
- System monitoring

Thank you.